

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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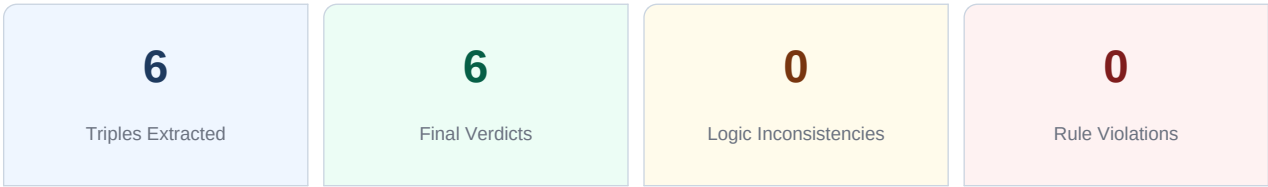
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REPORT OVERVIEW & METHODOLOGY

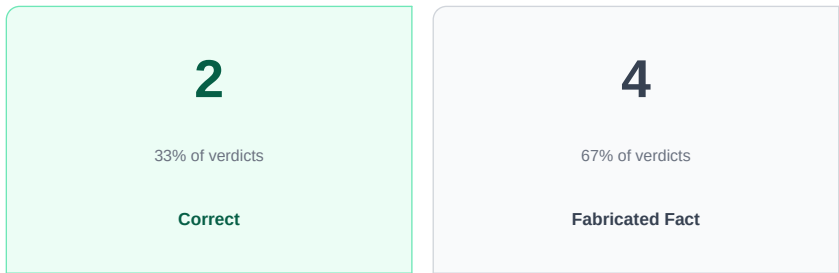
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	845	HAS_DEFAULT_PROBABILITY	not provided		Indicates that the default probability for credit score 845 is not specified in the data.
2	820	HAS_DEFAULT_PROBABILITY	not provided		Indicates that the default probability for credit score 820 is not specified in the data.
3	800	HAS_DEFAULT_PROBABILITY	0.7%		Specifies that the default probability for credit score 800 is 0.7%.
4	850	HAS_DEFAULT_PROBABILITY	0.4%		Specifies that the default probability for credit score 850 is 0.4%.
5	847	HAS_DEFAULT_PROBABILITY	estimated as slightly higher than 0.4%		Indicates that the default probability for 847 is estimated, based on the trend, as slightly higher than 0.4%.
6	847	HAS_DEFAULT_PROBABILITY	0.41%		Specifies the final estimated default probability for credit score 847, extrapolated to be 0.41%.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
845	HAS_DEFAULT_PROBABILITY	not provided	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH
820	HAS_DEFAULT_PROBABILITY	not provided	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH
800	HAS_DEFAULT_PROBABILITY	0.7%	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH
850	HAS_DEFAULT_PROBABILITY	0.4%	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH
847	HAS_DEFAULT_PROBABILITY	estimated as slightly higher than 0.4%	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH
847	HAS_DEFAULT_PROBABILITY	0.41%	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

No symbolic constraint violations detected.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
845	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	not provided	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	OBJECT	●●●●●●●● ●●● 1.00	DB candidate provides a concrete value ('0.7') while the triple asserts 'not provided', indicating fabrication of the object. Ev: subject: 845, predicate: DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE, object: 0.7
820	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	not provided	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	OBJECT	●●●●●●●● ●●● 1.00	DB candidate provides a concrete value ('0.7') while the triple asserts 'not provided', indicating fabrication of the object. Ev: subject: 820, predicate: DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE, object: 0.7
800	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	0.7%	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●■ 0.80	The triple object '0.7%' is a minor semantic difference from the DB candidate '0.7'; percentage notation does not change the value. Ev: subject: 800, predicate: DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE, object: 0.7

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
850	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	0.4%	Correct	factually_consistent	—	SEMANANTIC	—	●●●●●●●● ●■ 0.80	The triple object '0.4%' is a minor semantic difference from the DB candidate '0.4'; using a percentage sign does not alter the meaning. <i>Ev: subject: 850, predicate: DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE, object: 0.4</i>
847	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	estimated as slightly higher than 0.4%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● ●■ 0.90	The object asserts a concrete estimated value unsupported by any DB evidence for this subject and predicate.
847	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	0.41%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● ●■ 0.90	The object asserts a concrete value '0.41%' unsupported by any DB evidence for this subject and predicate.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence rule.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship that is in db but the subject, object or subject+object does not supported by db.